



University of Khartoum

Faculty of Mathematical Sciences and Informatics

GridGuard Khartoum

AI-Based Smart Load Shedding and Energy Management System for Developing Countries

A graduation project presented to the Faculty of Mathematical Sciences and Informatics, University of Khartoum, in partial fulfilment of the requirements for the degree of Bachelor of Computer Science.

Murtada Asim Ibrahim Bakhit (18-317)
Suliman Elawad Suliman Babiker (18-310)
Abdalhay Abass Mobarak Abass (15-306)

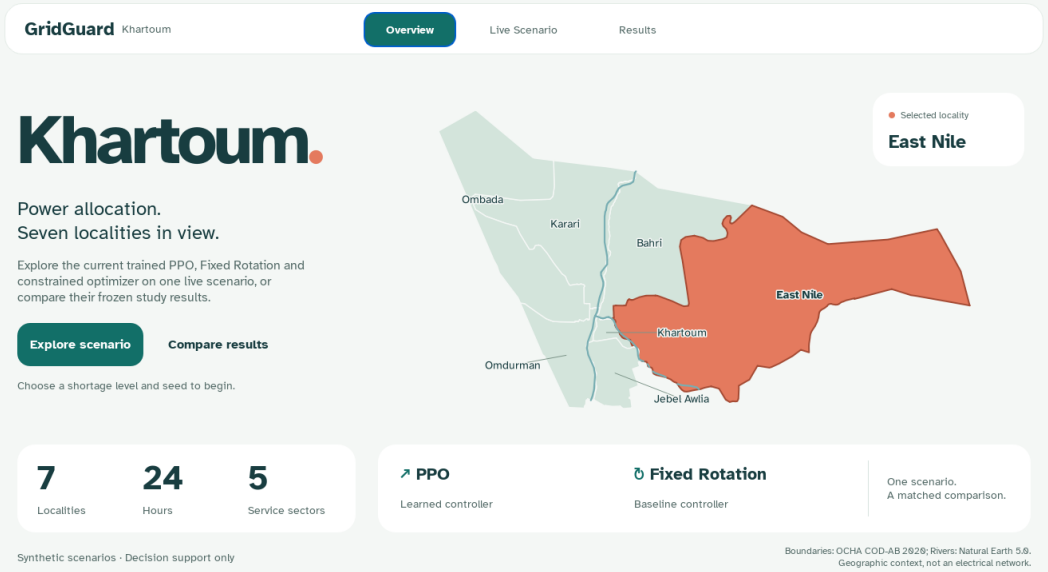
Project Supervisor: Mrs. Abubakr Siraj

October 2026

Hourly shortage allocation in Khartoum

Seven localities, five sectors and a shared supply pool

- Protect Hospital, Water and Bakery demand.
- Allocate remaining supply to Residential and Commercial demand.
- Measure service, interruption duration and distribution efficiency.



All demand and supply traces are synthetic. The map shows administrative geography.

Context: World Bank (2025). Related control research: Chen et al. (2023).

The simulation and service choices

24 hourly decisions with current observations and accumulated service history

Service tier	Requested demand
Off	No sectors
Critical Only	Hospital, Water and Bakery
Full Service	All five sectors

Shortage	Supply construction
Mild	80% of total demand
Moderate	65%, minus 10 pp at hours 17–20
Severe	50%, minus 15 pp at hours 16–21

- Independent demand and supply perturbations within ±5%.
- One supply pool. No network power flow, storage or restoration transients.

Critical protection through feasible actions

PPO and Constrained Optimization share 128 Critical Only / Full Service combinations

Critical reservation

$$C + \sum_{i=1}^{7} x_i b_i \le S, \qquad x_i \in \{0, 1\}$$

C: total critical demand b_i: locality noncritical demand S: available supply

$$\alpha = \min\left(1, \frac{S}{\max(Q, \varepsilon)}\right), \qquad \text{delivered} = \alpha \, \text{requested}$$

Shared proportional dispatch uses $\varepsilon = 10^{-8}$. If critical demand cannot fit, the all-Critical-Only fallback still records the resulting interruption.

Masking: Huang and Ontañón (2022). Safety distinction: García and Fernández (2015).

Three decision procedures

The optimizer and PPO receive equivalent causal information and the same feasible set

Controller	Decision rule	Horizon
Fixed Rotation	Three localities Full Service for each three-hour block	Fixed schedule
Constrained Optimization	Checks 128 assignments and maximizes current reward over the feasible set	Current hour
PPO	Chooses the highest-logit feasible action from a learned policy	Daily training objective

$$a_t^{\mathrm{opt}} = \arg\max_{a \in \mathcal{A}(s_t)} r_t(s_t, a), \qquad \pi^{\mathrm{PPO}}: \ \max \mathbb{E}\left[\sum_{t=0}^{23} r_t\right]$$

Reward and public service measures

The training objective combines protection, utilization, fairness and switching

$$r_t = \frac{5C_t + 2U_t + 2J_t - 0.25W_t - 3O_t}{9}$$

C: hourly critical service U: supply utilization J: cumulative-service fairness
W: fraction changing tier O: excess requested power / total demand

$$J_t = \frac{\left(\sum_{i=1}^{7} z_{i,t}\right)^2}{7\sum_{i=1}^{7} z_{i,t}^2 + \varepsilon}$$

z_i,t is cumulative served energy / cumulative demand. Report mean hourly Jain fairness, direct duration measures and energy separately from daily reward.

Fairness definition: Jain, Chiu and Hawe (1984).

Masked PPO training

Separate actor and critic networks: 46 inputs, two 256-unit tanh layers, 128 actor logits

$$\rho_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}$$

$$\mathcal{L} = -\mathbb{E}_t\left[\min\left(\rho_t \widehat{A}_t, \text{clip}(\rho_t, 0.8, 1.2)\widehat{A}_t\right)\right] + 0.25\mathbb{E}_t[(V_\phi - \widehat{R}_t)^2] - c_H\mathbb{E}_t[H(\pi_\theta)]$$

The same stored feasibility mask defines both policy probabilities.
Advantages normalize over the full rollout. Value targets remain fixed.

γ = 1.0, GAE λ = 0.95, zero terminal value. The critic term is exactly 0.25 MSE. Entropy coefficient decreases from 0.01 to 0.001.

PPO: Schulman et al. (2017). Masking: Huang and Ontañón (2022).

Five independent runs and frozen selection

Fresh initialization, disjoint training streams and a fixed training budget

Training element	Configuration
Training seeds	1729, 2718, 3141, 5772, 8111
Each run	3,000 iterations × 256 episodes × 24 hours
Total PPO exposure	92,160,000 transitions
Initialization	Fairness Rule demonstrations: 32,760 states, 10 epochs
PPO updates	4 epochs, minibatch 1,024, learning rate 0.0003 to zero

Select the best validation reward among checkpoints with zero critical/power violations and fairness/utilization at least Fixed Rotation overall and at every shortage level.

Matched evaluation and uncertainty

All final-test decisions follow the frozen five-run validation selection

Split	Independent base draws	Coupled conditions
Validation	5 seeds × 32 = 160	480
Final test	10 seeds × 32 = 320	960

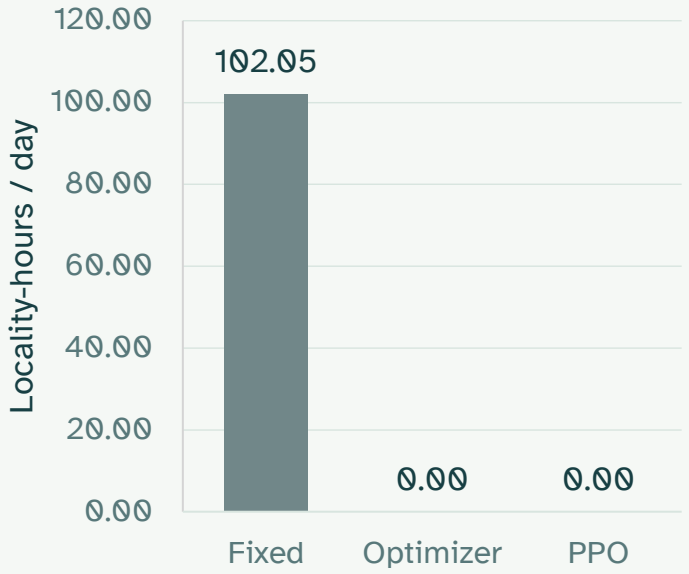
- 12 methods: Fixed Rotation, optimizer, five selected PPO actors and five iteration-600 actors.
- 11,520 trajectories. Bootstrap resamples five training runs and matched base-draw clusters.
- 10,000 resamples. Two primary claims use adjusted 97.5% intervals.

Statistical reliability: Agarwal et al. (2021).

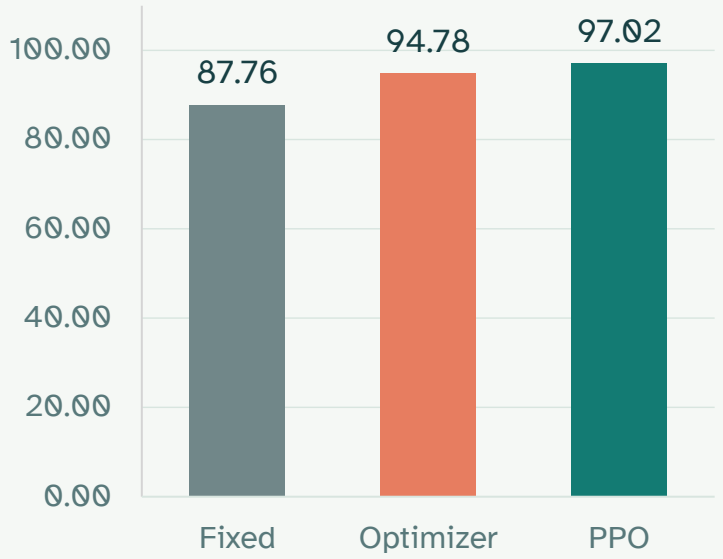
Main outcomes on 960 matched conditions

PPO values average the five independently trained policies equally

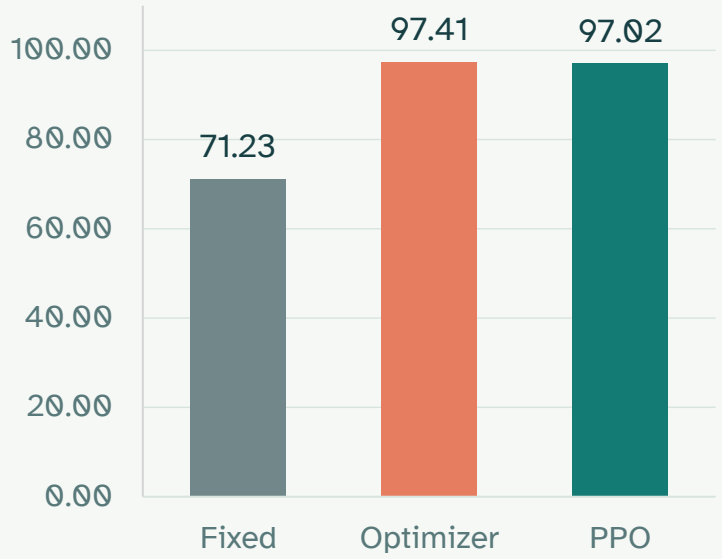
Critical interruption



Jain service fairness (%)



Supply utilization (%)



PPO and the optimizer preserve critical service. Their remaining differences concern allocation.

PPO improves service over Fixed Rotation

Paired effects are PPO minus Fixed Rotation. Intervals have adjusted 97.5% coverage.

Outcome	Effect [lower, upper]
Critical interruption (locality-hours)	-102.050 [-102.081, -102.025]
Jain fairness (percentage points)	+9.260 [+9.142, +9.371]
Utilization (percentage points)	+25.782 [+25.702, +25.859]

Full declared comparison: PASSED

This compares the complete constrained system with a supply-blind fixed schedule. The critical-service rule contributes to the improvement.

A reward and fairness gain with a utilization tradeoff

Paired effects are PPO minus Constrained Optimization. Adjusted 97.5% intervals.

Outcome	Effect [lower, upper]
Daily reward	+0.0499 [+0.0406, +0.0577]
Jain fairness (percentage points)	+2.237 [+2.096, +2.373]
Utilization (percentage points)	-0.399 [-0.464, -0.341]
Switching (percentage points)	+7.218 [+6.531, +8.112]

Full service-guarded improvement criterion: NOT MET

The utilization loss exceeds the allowed 0.2-point margin. Both protect critical service.

Duration fairness and total interruption burden

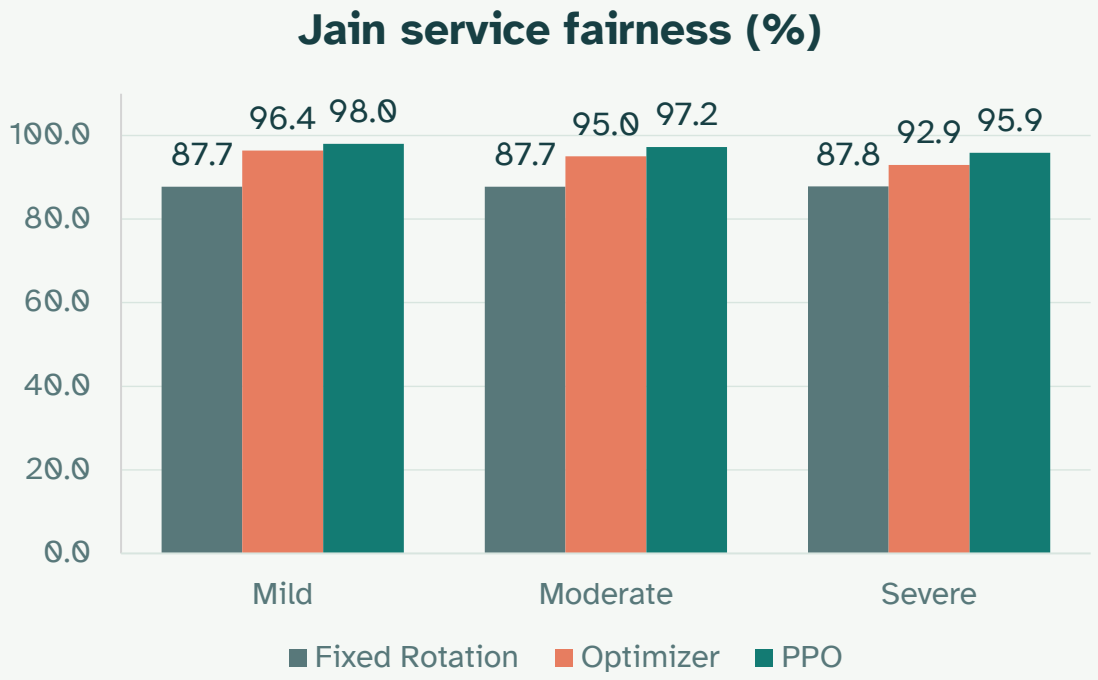
Episode duration measures are calculated before averaging

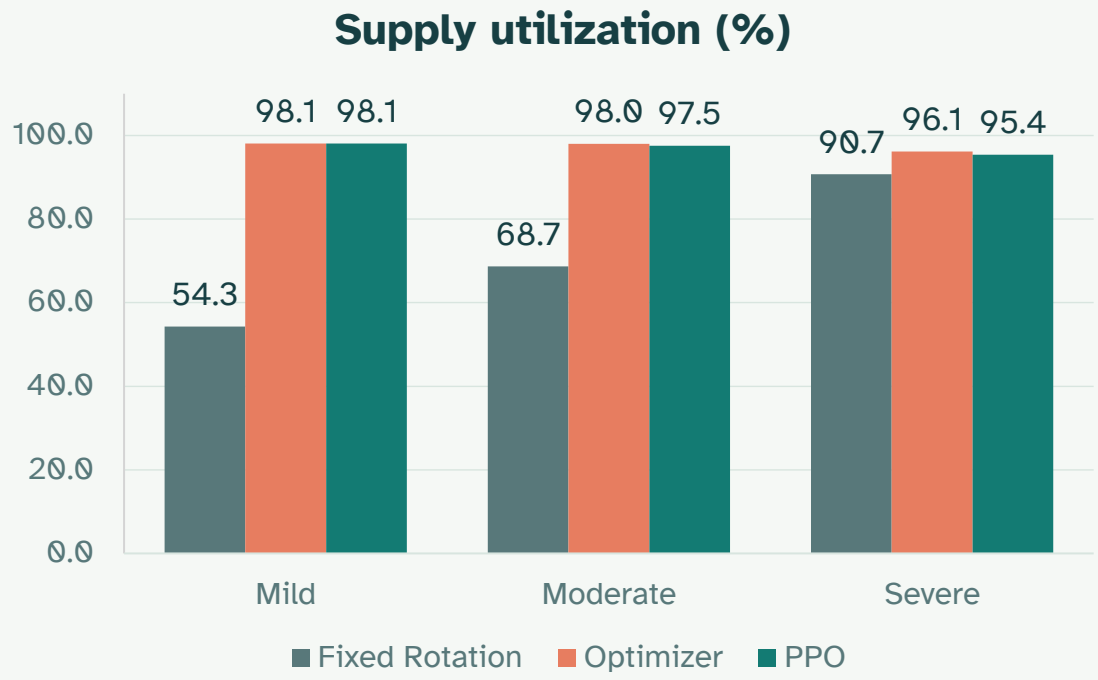
Measure	Fixed Rotation	Optimizer	PPO
Zero-service blackout (locality-hours)	96.00	0.00	0.00
Blackout pairwise gap (hours)	1.71	0.00	0.00
Critical-interruption pairwise gap (hours)	1.90	0.00	0.00
Longest critical interruption (hours)	8.00	0.00	0.00

Critical Only removes locality-wide zero-service blackouts while Residential and Commercial demand can remain unmet. Zero duration does not mean Full Service.

Shortage-specific allocation outcomes

Results keep the Mild, Moderate and Severe conditions visible

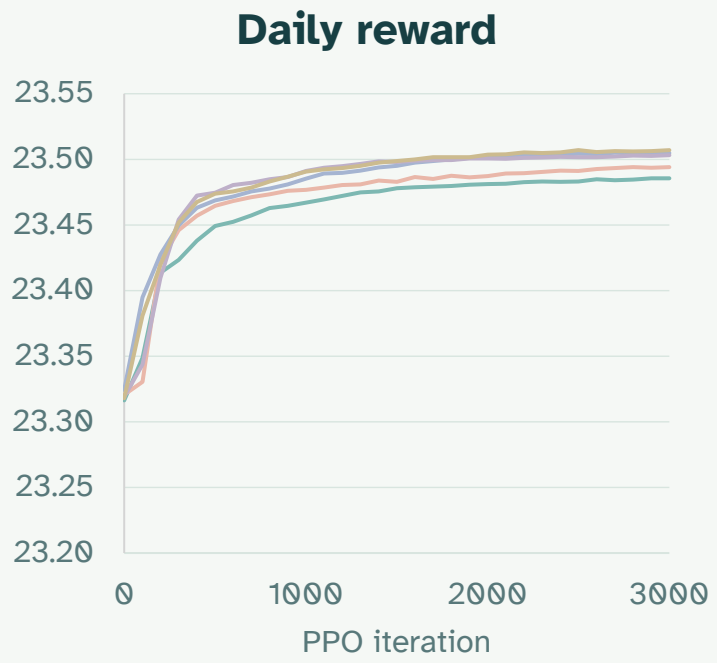


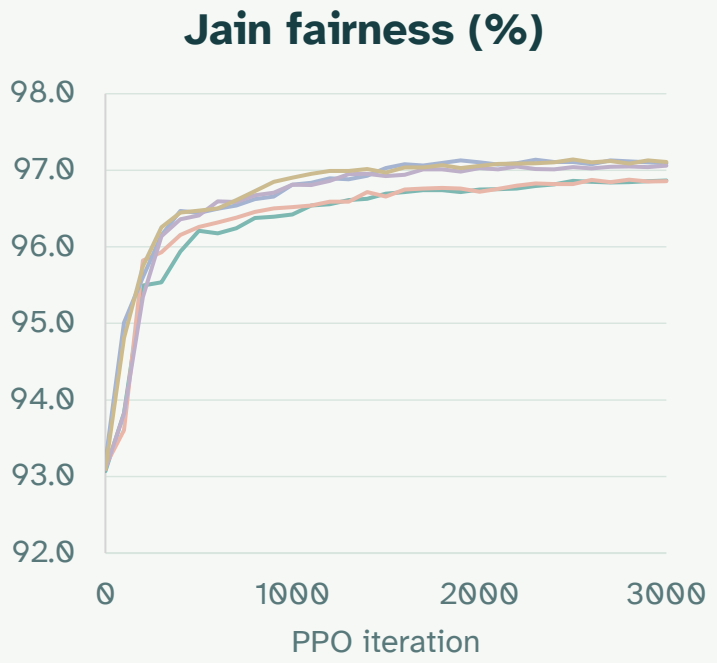


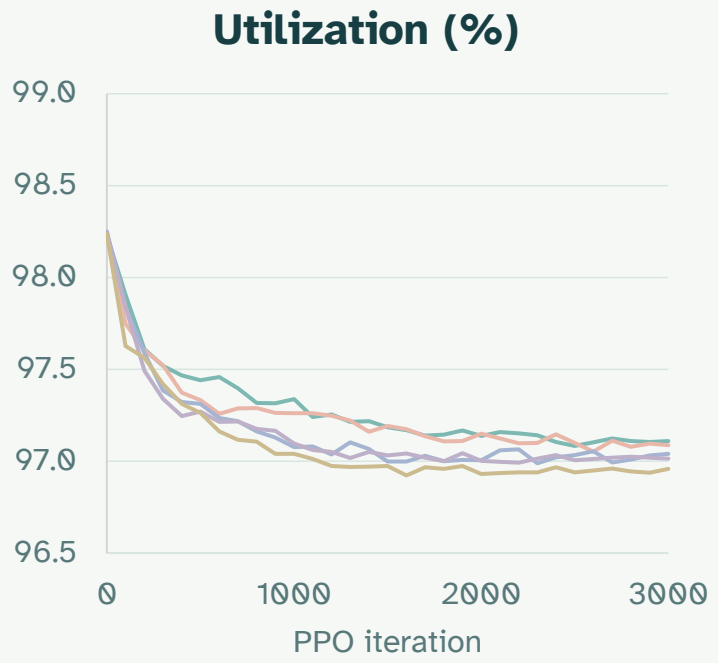
Critical interruption is zero for PPO and the optimizer at every tested shortage level.

Validation progress across all five training runs

31 checkpoints per run. Checkpoint selection never consults final-test performance.



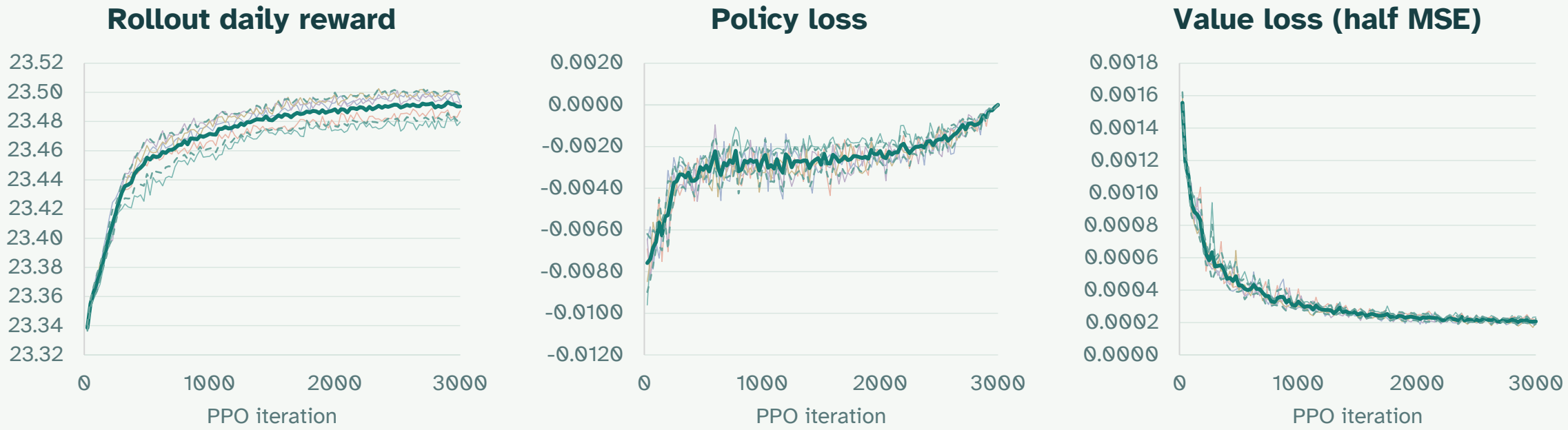




Seeds: 1729 2718 3141 5772 8111

Training reward and loss

120 recorded updates per seed, every 25 iterations. No smoothing.



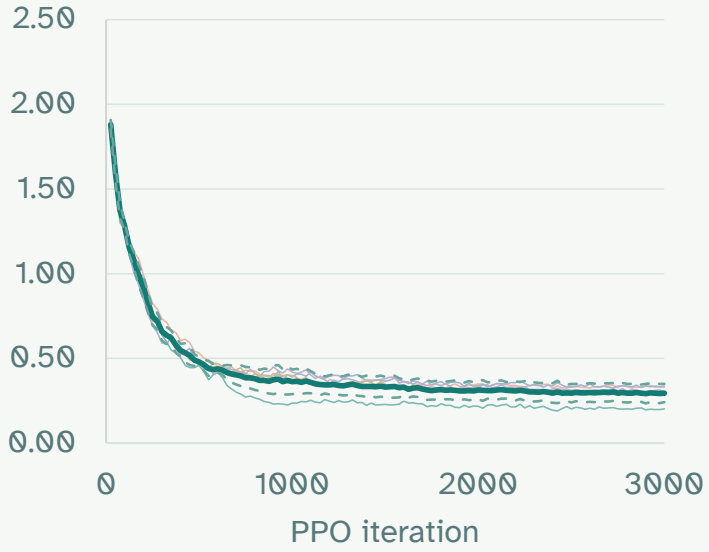
Thin lines: five seeds. Dark line: mean. Dashed limits: mean ± sample SD, not a CI.

Reward describes the pre-update rollout. Losses average all minibatches of that update’s four epochs.

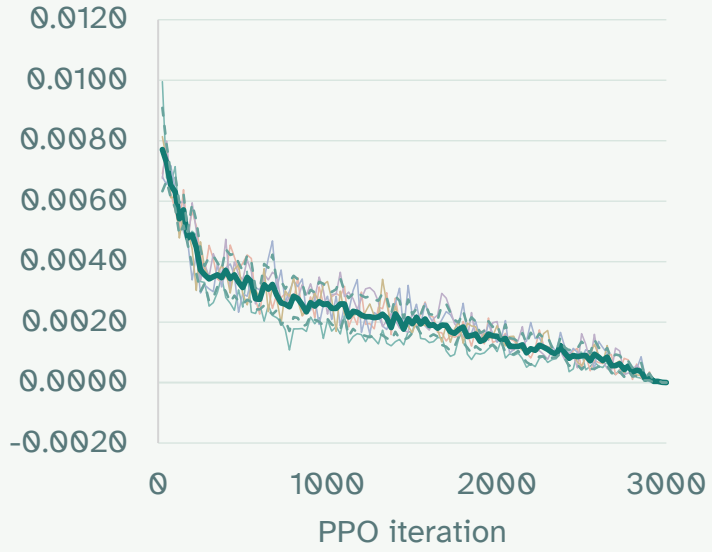
Policy entropy and update diagnostics

120 recorded updates per seed, every 25 iterations. No smoothing.

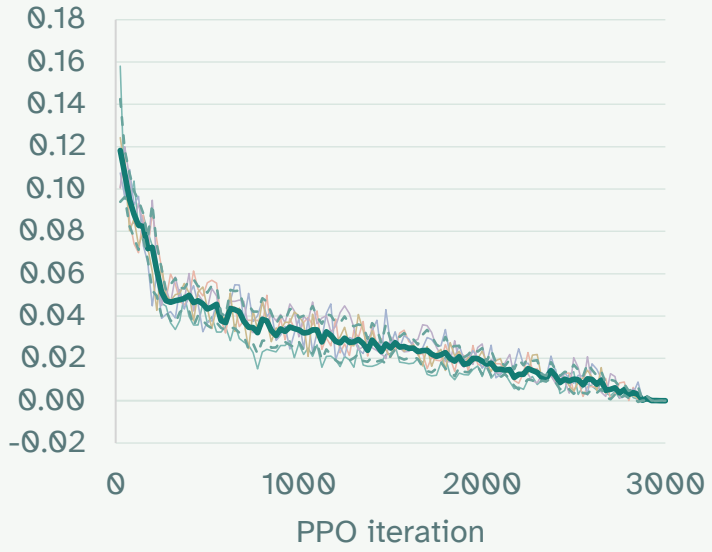
Policy entropy (nats)



Approximate KL



Clipped fraction



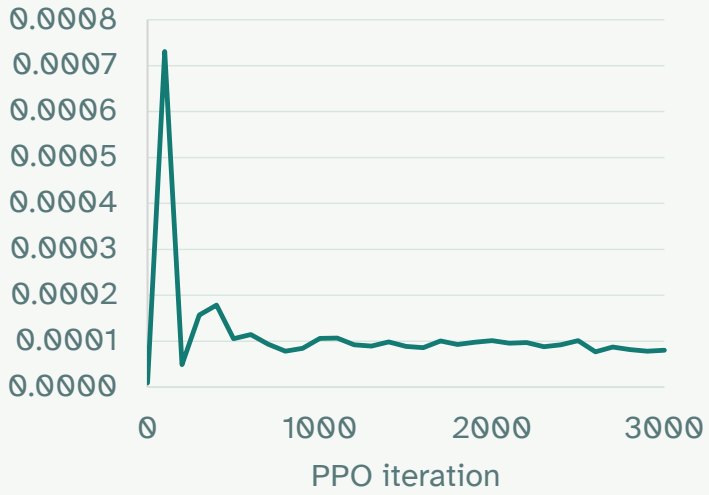
Thin lines: five seeds. Dark line: mean. Dashed limits: mean ± sample SD, not a CI.

Entropy, KL and clipping average all minibatches of the logged update. They are not 25-update moving averages.

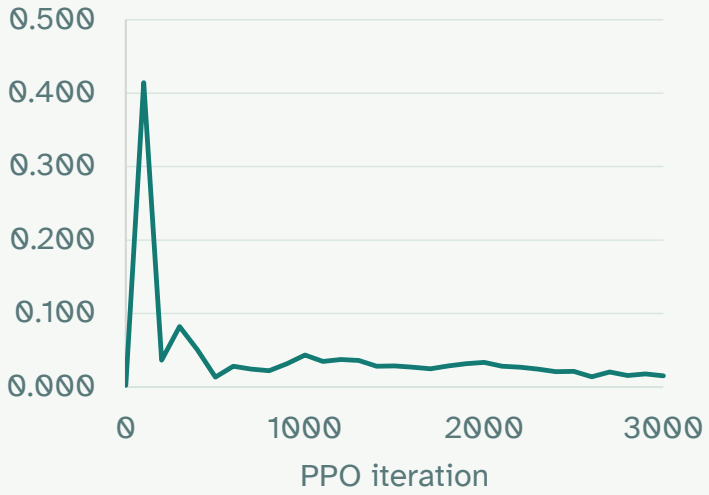
Between-seed validation variance

Sample variance of the five run means on the same validation scenarios

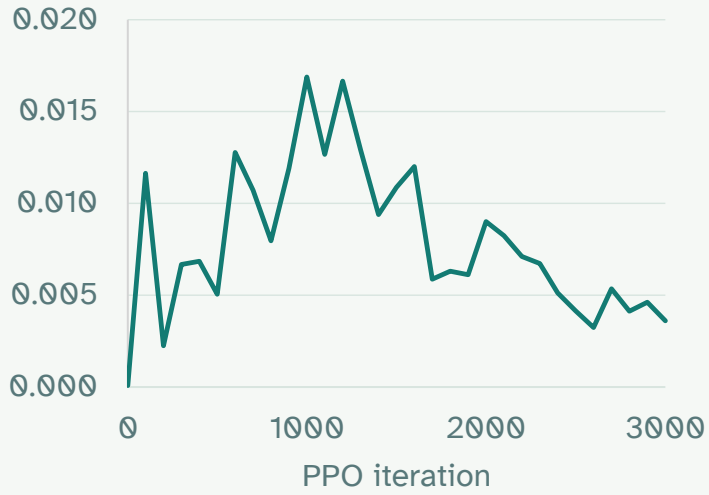
Reward variance (score²)



Fairness variance (pp²)



Utilization variance (pp²)



$$\bar{x}_j = \frac{1}{5}\sum_{k=1}^{5} x_{k,j}, \qquad s_j^2 = \frac{1}{4}\sum_{k=1}^{5}(x_{k,j} - \bar{x}_j)^2$$

31 checkpoints include behavior-cloning iteration 0. This is between-run variance, not critic explained variance or final-test uncertainty.

Longer training improves reward but loses utilization

Same-run iteration-600 snapshots are a within-schedule reference

Seed	Selected iteration	Reward at 600	Selected reward	Difference
1729	2900	23.4497	23.4835	+0.0338
2718	3000	23.4654	23.4913	+0.0258
3141	2900	23.4699	23.5034	+0.0335
5772	3000	23.4781	23.5014	+0.0233
8111	2500	23.4732	23.5040	+0.0309

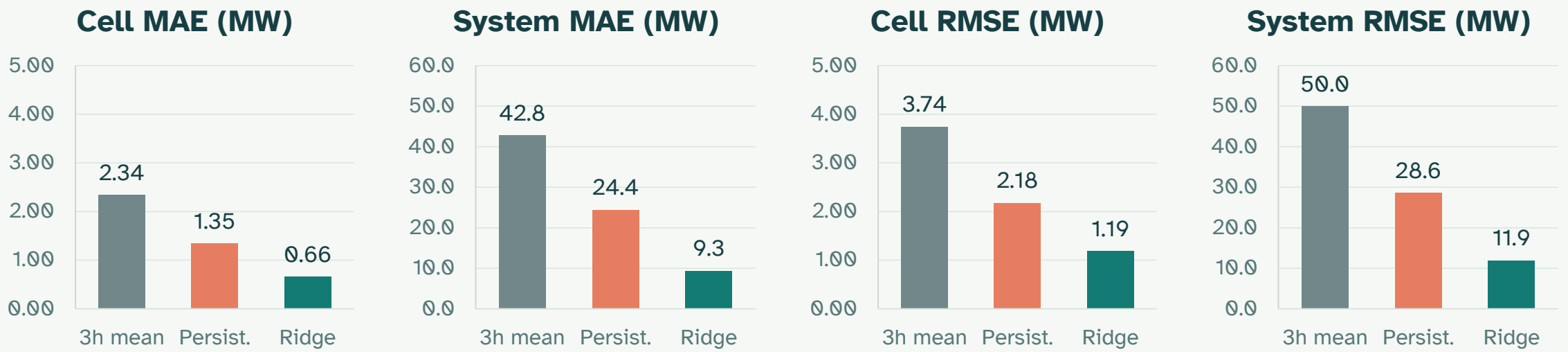
Mean daily reward +0.0295. Utilization −0.2417 percentage points.

The full additional-training criterion fails its utilization and shortage guardrails. The 600 snapshots use the planned 3,000-iteration schedule.

Learned next-hour demand prediction

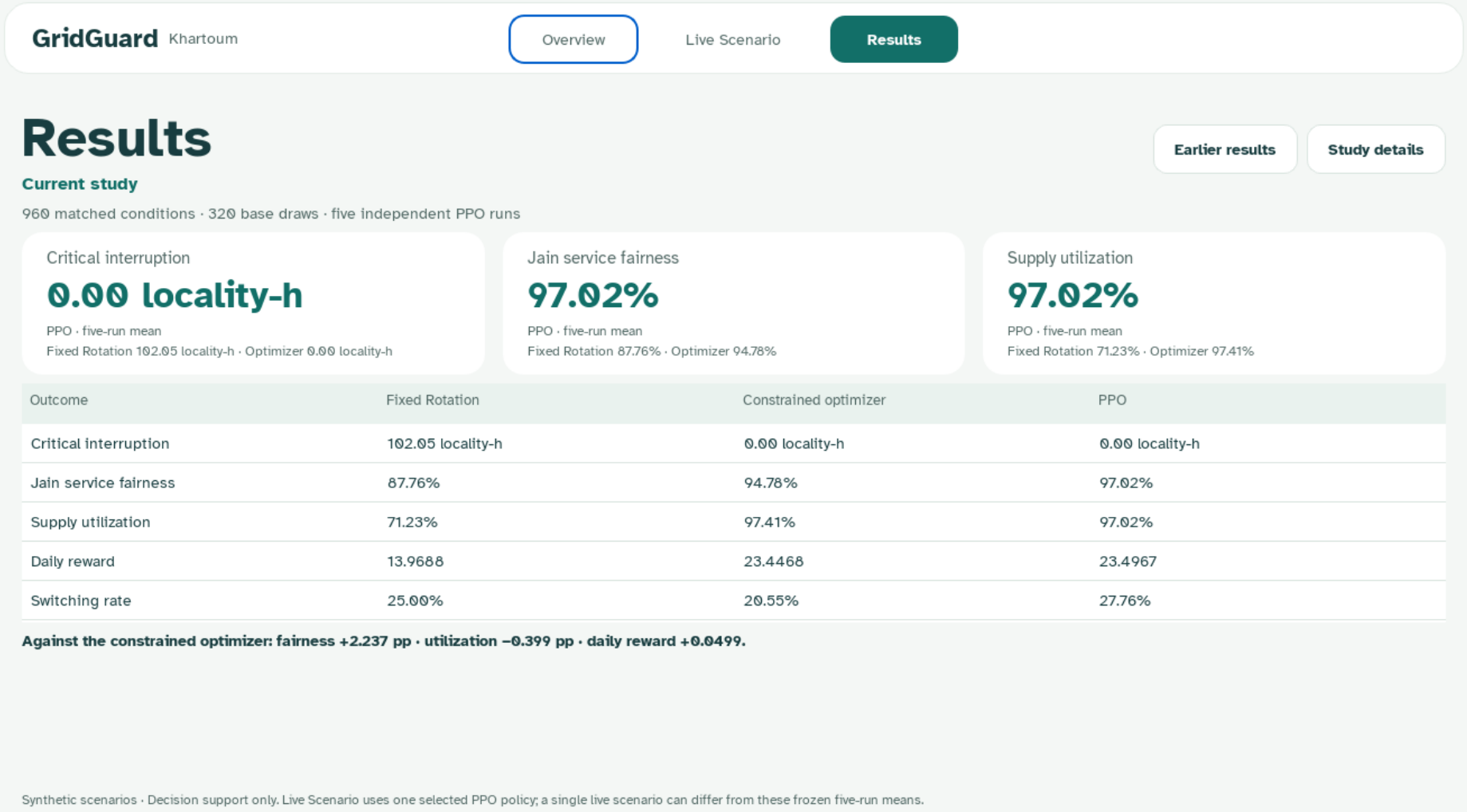
Ridge AR(3): 768 training traces and 320 separate evaluation traces

$$\hat{d}_t = \max(0, \beta_0 + \beta_1 d_{t-1} + \beta_2 d_{t-2} + \beta_3 d_{t-3})$$



Forecasts use past demand and identical target hours. Prediction does not enter PPO.

Current study results in the desktop app



Outcome	Fixed Rotation	Constrained optimizer	PPO
Critical interruption	102.05 locality-h	0.00 locality-h	0.00 locality-h
Jain service fairness	87.76%	94.78%	97.02%
Supply utilization	71.23%	97.41%	97.02%
Daily reward	13.9688	23.4468	23.4967
Switching rate	25.00%	20.55%	27.76%

Conclusions and the next engineering step

Against Fixed Rotation

The constrained PPO system improves critical service, fairness and utilization on fresh matched synthetic scenarios.

Against the optimizer

PPO improves fairness and daily reward but uses less supply and switches more often. The full criterion fails.

- Ridge AR(3) improves prediction over the two simple baselines.
- Earlier Fairness Rule comparisons remain unfavorable for PPO.
- Measured demand and feasible feeder actions are needed before operational evaluation.

References 1 of 2

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